Solution to MP363 Problem Set 6

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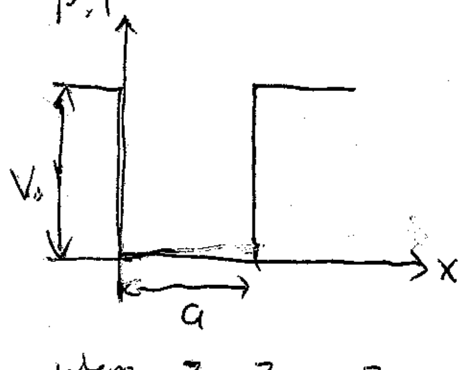


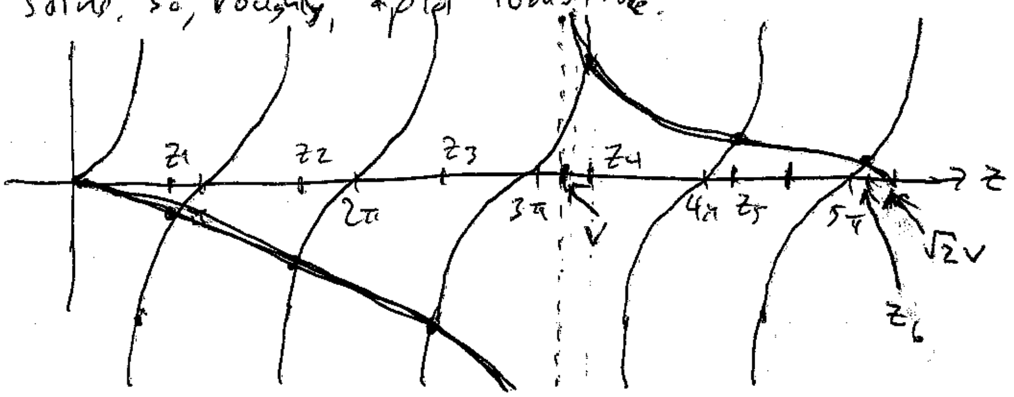
In lecture, we showed that the energies of the bound states for a finite potential well of depth V_0 and width a (as shown to the left) are

$$E_n = \frac{\hbar^2 z_n^2}{2ma^2}$$

where $z_1, z_2, \dots, z_N$ are the solutions to the equation

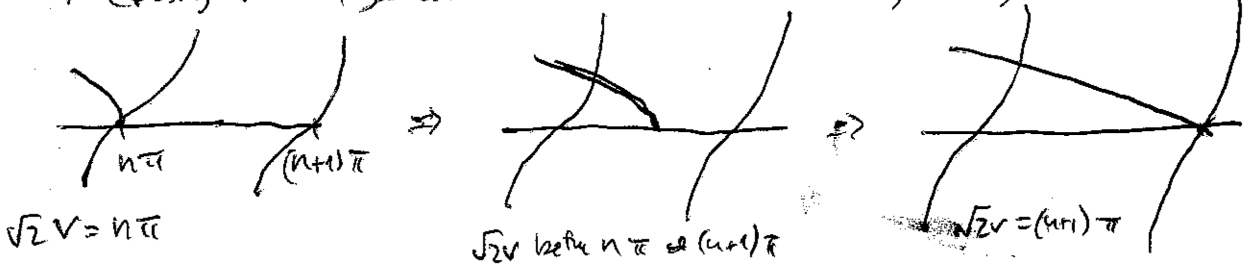
$$\tan z = \frac{z\sqrt{2v^2 - z^2}}{z^2 - v^2}$$

where $v = \sqrt{ma^2 V_0^2/\hbar^2}$ and the solutions are ordered so that $z_1 < z_2 < \dots < z_N$. The question is, how many such solutions are there? Graphically, we can get a rough idea of this by plotting $\tan z$ and $\frac{z\sqrt{2v^2-z^2}}{z^2-v^2}$ as functions of z on the same set of axes. $\tan z$ is a well-known function, and we can get an idea of what the other looks like: at $z=0$, $\frac{z\sqrt{2v^2-z^2}}{z^2-v^2} = 0$, obviously. For $z < v$, it's negative, blowing up (down?) to $-\infty$ as $z \to v$ from the left. For $z \to v$ from the right, it's positive and infinite. It falls to zero again at $z = \sqrt{2}\,v$, and so $v < z < \sqrt{2}\,v$ gives a positive value. For $z > \sqrt{2}\,v$, it's imaginary, and so we know there will be no solutions. So, roughly, a plot looks like:



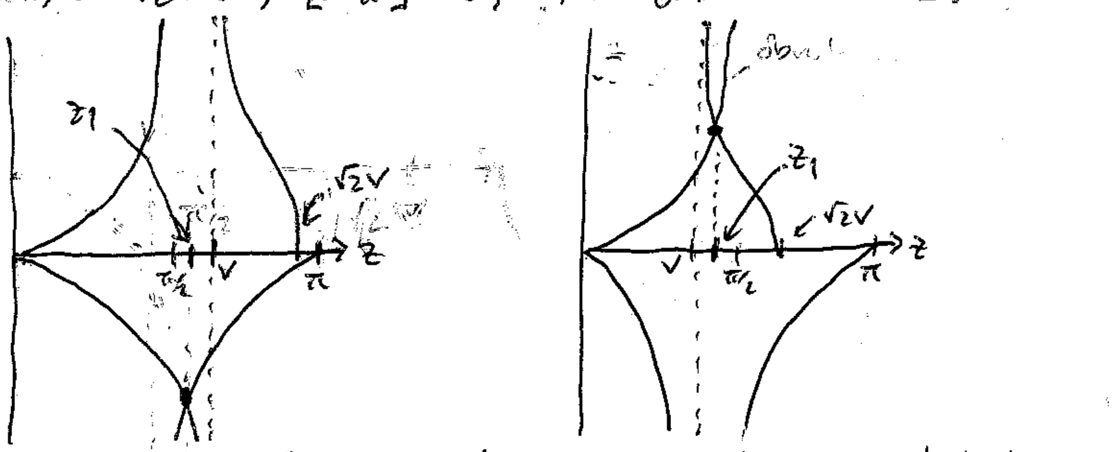
(This would be for v between 3π and 4π.) For this, we see $N = 6$, i.e. there are 6 values of z where the plots intersect. One way to count the total number of crossings goes like this: $\sqrt{2}\,v$ is the largest value z can possibly be, and as we approach it from the left, $\frac{z\sqrt{2v^2-z^2}}{z^2-v^2}$ is positive. Thus, it can only cross one of the "lobes" of $\tan z$ somewhere between $n\pi$ and $n\pi + \pi/2$ for n a positive integer (because $\tan z > 0$ for these values).

Now, if $\sqrt{2}v$ is exactly a multiple of π, then the final crossing is at $z=\sqrt{2}v=n\pi$. If $\sqrt{2}v$ is slightly larger than this, then the crossing is no longer at $n\pi$ but a bit to the right; however, if



$\sqrt{2}v<(n+1)\pi$, this is still the last intersection. It's only when $\sqrt{2}V=(n+1)\pi$ that we get the next intersection. Now, this means that N, the total number of crossings, must depend on v in such a way that as $\sqrt{2}V$ increases, N changes by 1 only when $\sqrt{2}V$ goes from the region between $n\pi$ to $(n+1)\pi$ or above.

Here's where the floor function comes in: $\sqrt{2}V/\pi$ changes continuously from n to $n+1$ in the process described above, but $\left\lfloor\frac{\sqrt{2}V}{\pi}\right\rfloor$ doesn't: for $n\pi\le\sqrt{2}V<(n+1)\pi$, $\left\lfloor\frac{\sqrt{2}V}{\pi}\right\rfloor=n$, but as soon as $\sqrt{2}v=(n+1)\pi$, $\left\lfloor\frac{\sqrt{2}V}{\pi}\right\rfloor$ jumps to $n+1$. So the number of crossings increases by 1 every time $\left\lfloor\frac{\sqrt{2}V}{\pi}\right\rfloor$ increases by 1. This tells us that $N=\left\lfloor\frac{\sqrt{2}V}{\pi}\right\rfloor+N_0$, where N_0 is an integer. But what's N_0? It's the smallest number of crossings we can have. Now by assumption, $V_0>0$, so $v>0$. Thus, $\frac{\sqrt{2}V}{\pi}>0$; however, as long as $\sqrt{2}v<\pi$, $\left\lfloor\frac{\sqrt{2}V}{\pi}\right\rfloor=0$, so $N=N_0$ for $0<v<\pi/\sqrt{2}$. How



many crossings are there? The answer: 1. The above two plots take into account both the cases that $\frac{\pi}{2}<v<\frac{\pi}{\sqrt{2}}$ and $v<\pi/2$. In the first case, the crossing is in the lower half-plane, and in the second, the upper half-plane, but in either case, there's only a single crossing. Thus, $N_0=1$, and we conclude that

$$\boxed{N = \left\lfloor \frac{\sqrt{2}V}{\pi} \right\rfloor + 1}$$ is the number of crossings, and thus, the number of bound states.

P.2

If $V_0 = 6\hbar^2/ma^2$, then $V = \sqrt{\frac{ma^2V_0}{\hbar^2}} = \sqrt{6}$, so our eq'n giving the bound state energies is $\tan z = \frac{z\sqrt{12-z^2}}{z^2-6}$.

(a) $\sqrt{2}V = \sqrt{12} = 3.464$, so $\frac{\sqrt{2}V}{\pi} = 1.103$. $\lfloor\frac{\sqrt{2}V}{\pi}\rfloor$ is the largest integer less than 1.103, which is 1. Therefore, we expect there to be $N = \lfloor\frac{\sqrt{2}V}{\pi}\rfloor + 1 = \boxed{2}$ bound states for this particular V_0.

(b) Ah, but what are the energies E_1 and E_2 of these two states? At the end of these solutions is a plot of both $\tan z$ (dotted line) and $\frac{z\sqrt{12-z^2}}{z^2-6}$ (solid line) done with the mathematical program Maple, and we can see precisely where the two intersect and read off the values of z_1 and z_2 from those. z_1 is a tiny bit less than 2. We can be more precise: there are 35mm between 1 and 2 on the plot, and z_1 is about 2.5mm to the left of 2, giving $z_1 \approx 2 - \frac{2.5}{35} = 1.93$. Similarly, z_2 is about 14.5mm to the right of 3, giving $z_2 \approx 3 + \frac{14.5}{35} = 3.41$.

But we can be a bit more exact: there's a well-known method for finding the value of z satisfying $f(z) = 0$ for some function f, called Newton's method, which says that if z_0 is an initial guess at one of the zeroes of f, then $z_1 = z_0 - \frac{f(z_0)}{f'(z_0)}$ is a better approximation, and $z_2 = z_1 - \frac{f(z_1)}{f'(z_1)}$ is better still. Put more precisely, if z_0 is an initial guess and we construct the sequence $\{z_0, z_1, z_2, \ldots\}$ given by

$$z_{n+1} = z_n - \frac{f(z_n)}{f'(z_n)}, \quad n = 0, 1, 2, \ldots$$

then z_n converges to a zero of f as $n \to \infty$. However, some functions have more than one zero, so we'll have to pick different initial guesses to get each zero. Unsurprisingly, the closer the initial guess is to a zero, the more quickly z_n will converge to the "real" zero.

So, I used this method in a spreadsheet (LibreOffice Calc, but Excel or something like it will also do the trick) with an initial guess of $z_0 = 2$, since the plot shows that one of the zeroes of

$$f(z) = \tan z - \frac{z\sqrt{12-z^2}}{z^2-6}$$

is near that. You can show that $f'(z) = \sec^2 z - \frac{72}{(z^2-6)^2\sqrt{12-z^2}}$, and so using Newton's method gives 1.9475 as a zero after about 10 iterations (not too far from the 1.93 I got by eyeballing the plot). Taking this thus means that the ground state energy of this system is

$$E_1 = \frac{\hbar^2}{2ma^2} z_1^2 = \frac{(1.9475)^2}{2} \cdot \frac{\hbar^2}{ma^2} = \boxed{1.8964\ \hbar^2/ma^2}$$

(or, if you like, $0.3161\, V_0$). The other zero is just below where $\frac{z\sqrt{12-z^2}}{z^2-6}$ falls to zero, or $\sqrt{12}$. Taking this as my initial guess for Newton's method, a few iterations gives 3.4285 as a zero of $f(z)$. Thus, the other bound state energy is

$$E_2 = \frac{\hbar^2}{2ma^2} z_2^2 = \frac{(3.4285)^2}{2} \cdot \frac{\hbar^2}{ma^2} = \boxed{5.8774\ \hbar^2/ma^2}$$

(or $0.9796\, V_0$). Not quite as nice as $E_n = \frac{\pi^2 \hbar^2}{2ma^2} n^2$ like the infinite square well, but this shows that we can still get the energies.

P.3

(a) Now for a step potential, and height $V_0 = 2eV$. We want to find how much of the beam of electrons is reflected if each electron has $E = 1eV$ of kinetic energy. In lecture, we derived

$$R = \frac{(k^2+K^2)^2 \sinh^2 Ka}{(k^2+K^2)^2 \sinh^2 Ka + 4k^2K^2} \qquad k = \sqrt{\frac{2mE}{\hbar^2}},\ K = \sqrt{\frac{2m(V_0-E)}{\hbar^2}}$$

but it's easier to write this directly in terms of V_0, E, m and $\hbar$, especially since $k^2+K^2 = \frac{2mV_0}{\hbar^2}$. This, and a little algebra, gives

$$R = \frac{V_0^2 \sinh^2\left(\sqrt{\frac{2ma^2(V_0-E)}{\hbar^2}}\right)}{V_0^2 \sinh^2\left(\sqrt{\frac{2ma^2(V_0-E)}{\hbar^2}}\right) + 4E(V_0-E)}$$

Now, if $V_0 = 2eV$ and $E = 1eV$, this reduces to

$$R = \frac{\sinh^2 Ka}{\sinh^2 Ka + 1}$$

We're told that $R = 0.9$; thus,

$$\frac{\sinh^2 Ka}{\sinh^2 Ka + 1} = 0.9 \Rightarrow 10\sinh^2 Ka = 9\sinh^2 Ka + 9 \Rightarrow \sinh Ka = 3$$

and so

$$a = \frac{1}{K}\sinh^{-1}(3).$$

Putting in the numbers gives

$$a = \sqrt{\frac{\hbar^2}{2m(V_0-E)}}\sinh^{-1}(3) = \sqrt{\frac{(1.05\times10^{-34}\,J.s)^2}{2(9.11\times10^{-31}\,kg)(1.6\times10^{-19}\,J)}}\ (1.818)$$

$$= \boxed{3.535\times10^{-10}\,m = 3.535\ \text{Å}}$$

which is about the size of a large atom.

(b) We now have to be careful for this case, because E is now the same as V_0. However, as $E \to V_0$, we see both the numerator and denominator of R go to zero, so we need to use some method to find R in this limit. l'Hôpital's rule will work, but we can also use Taylor series: $\sinh x = \frac{e^x - e^{-x}}{2}$, and since $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$ and $e^{-x} = 1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \dots$

we have

$$\sinh x = x + \frac{x^3}{6} + \frac{x^5}{120} + \dots = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}.$$

If E is near V_0, then $E - V_0$ is small, so κa is small, giving

$$\sinh^2 \kappa a \approx (\kappa a)^2 = \frac{2ma^2}{\hbar^2}(V_0 - E).$$

Thus, for E near V_0, $\lim_{E \to V_0} R$ is

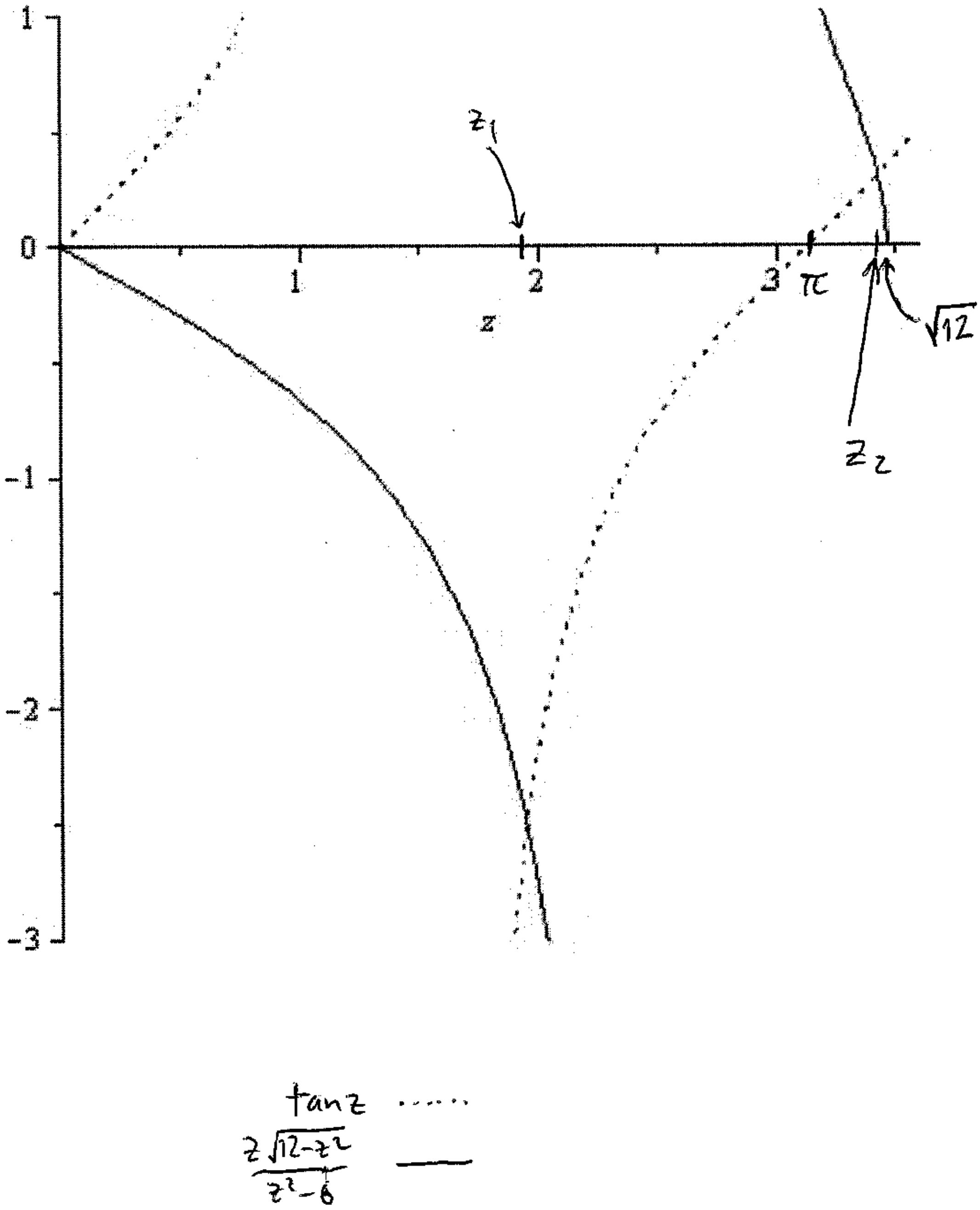
$$\lim_{E \to V_0} \frac{V_0^2 \frac{2ma^2}{\hbar^2}(V_0 - E)}{V_0^2 \frac{2ma^2}{\hbar^2}(V_0 - E) + 4E(V_0 - E)} = \lim_{E \to V_0} \frac{2ma^2 V_0^2/\hbar^2}{\frac{2ma^2 V_0^2}{\hbar^2} + 4E} = \frac{V_0}{V_0 + 2\hbar^2/ma^2}$$

and so if the electron beam energy is the same as the barrier height, we have

$$R = \frac{2 \times 1.6 \times 10^{-19}\,\mathrm{J}}{2 \times 1.6 \times 10^{-19}\,\mathrm{J} + 2(1.05 \times 10^{-34}\,\mathrm{J \cdot s})^2 / (9.11 \times 10^{-31}\,\mathrm{kg})(3.575 \times 10^{-10}\,\mathrm{m})^2}$$

$$= \boxed{0.623}$$

so about 62% of the beam bounces off the barrier and about 38% tunnels through it.


1
0
-1
-2
-3
1
2
3
z
z_1
π
$\sqrt{12}$
z_2
$\tan z$ ......
$\frac{z\sqrt{12-z^2}}{z^2-6}$ ——